

Sub 6 GHz versus mmWave Measurements in a Controlled High-Mobility Environment

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Abstract— Millimeter-wave (mmWave) high-mobility scenarios are deployment scenarios for future wireless communication systems. Measurements in such scenarios cannot be easily repeated, because outdoor environments are hardly controllable. Furthermore, such high-mobility scenarios pose challenges for performance measurements as such measurements are expensive, complex, and time-consuming. Consequently, comparing consecutive measurements in different bands is not straightforward. This paper presents a methodology to ensure a fair comparison between sub 6 GHz and mmWave systems by conducting repeatable, closed-loop, and off-line-processed measurements in a controlled indoor environment. The design is based on a rotating arm, spinning a lightweight transmitter chip with an antenna around a central pivot. Thereby, such high-velocity experiments can be performed at the same channel conditions with low cost and complexity in a repeatable manner.

Keywords— mmWave, sub 6 GHz, 5G, vehicular communications, high-speed train, channel measurements

I. INTRODUCTION

In the last decade, the demand for high data rates in high-mobility scenarios, such as vehicular and high-speed train (HST) scenarios, has severely increased. Such high data rates require a large bandwidth, which cannot be supported by conventional 3G and 4G technologies operating in the sub 6 GHz bands. Compared with sub 6 GHz bands, millimeter wave (mmWave) bands (10 GHz – 300 GHz) provide extremely large signal bandwidths [1]. For this reason, future wireless communication systems consider mmWave communication as one of their core technologies.

Sub 6 GHz high-mobility scenarios have been well investigated in realistic environments [2]–[4], as well as in controllable measurement environments [5]–[9]. Besides sub 6 GHz bands, measurements were also performed at mmWave frequencies in real-world, highly time-variant radio channels [10]–[18]. Several comparisons between sub 6 GHz and mmWave bands have been derived through simulations [19]–[21] and measurements [22]. The problem here is when using different antennas for different frequency bands, these antennas cannot be placed at the same position at the same time. If we want to measure at the same position, we need to measure at different times, therefore we need repeatability.

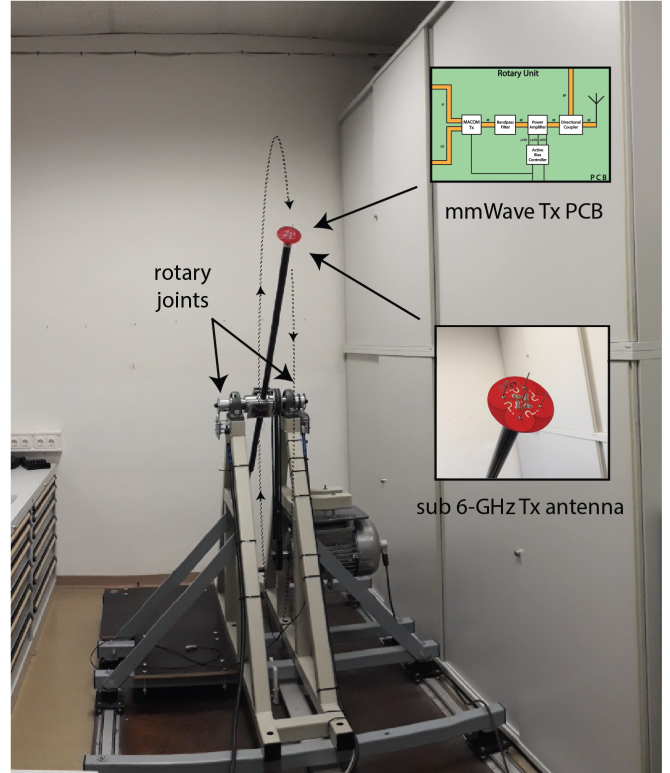


Fig. 1. Measurement setup to compare sub 6 GHz and mmWave in high-mobility environment. There is either a sub 6 GHz Tx antenna or a mmWave Tx PCB mounted at the end of the rotary arm.

Contribution: This paper proposes a methodology to enable a fair comparison between sub 6 GHz and mmWave bands in a repeatable¹ and controlled² high-mobility environment. Through repeatability, we are able to compare measurements at different bands and different times conducted at the same position with each other. Therefore, we enable comparisons of wireless channels at different bands in terms of small scale and fast-fading.

¹Repeatability is the ability to repeat a measurement with equal channel conditions, which is hardly possible by conducting real-world drive-by measurements.

²Controllability means being able to control environmental as well as system parameters, for example, the signal-to-noise ratio (SNR).

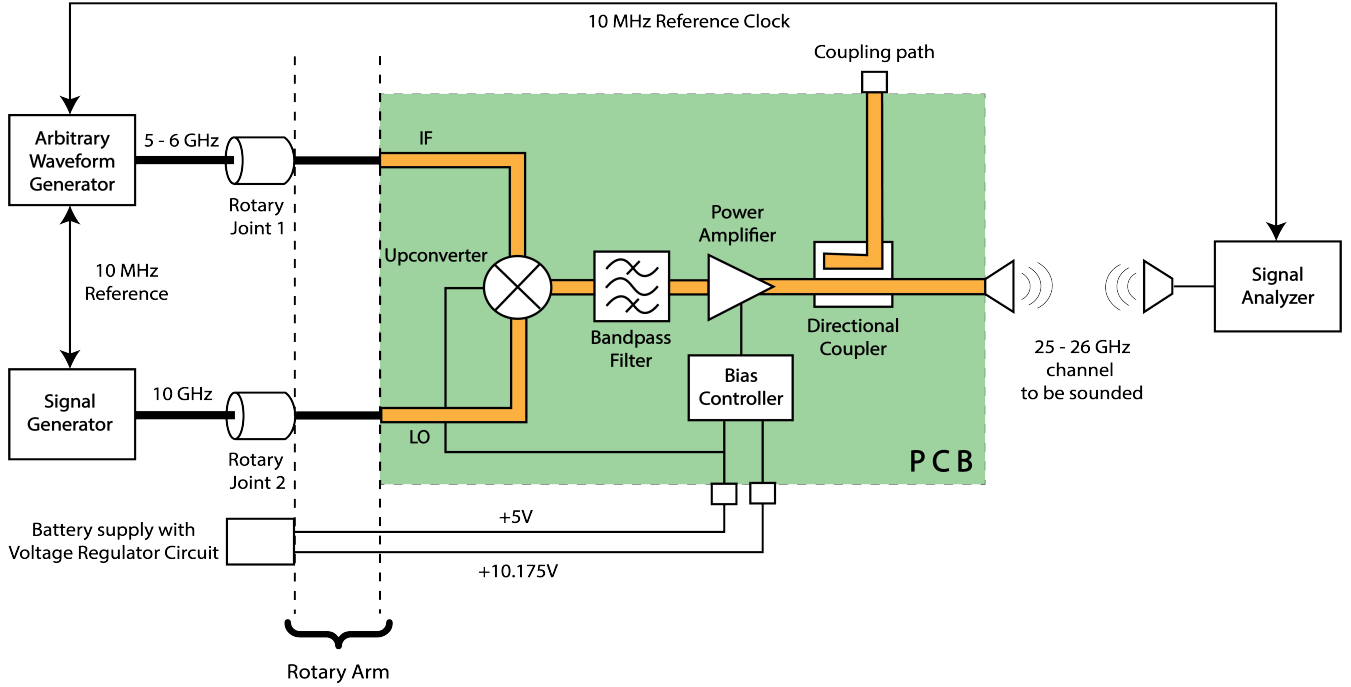


Fig. 2. Measurement setup employed to sound a mmWave channel. We mount a mmWave transmitter PCB at the end of the rotary arm.

II. TESTBED ARCHITECTURE

In this section, we provide a detailed description of the proposed testbed. Our testbed's current design is based on a rotary unit that rotates an antenna around a central pivot with a constant speed of up to 400 km/h (see Fig. 1) [23], [24]. The transmit signal is fed through a rotary joint and then through the rotating arm to the antenna. This allows for the transmission of sub 6 GHz signals.

The rotary joint has an upper-frequency limit of 18 GHz and a maximum continuous rotational speed of 1500 rpm. Therefore, our testbed's current design does not allow transmitting a mmWave signal through the rotary joint directly to the end of the rotary arm. To our knowledge, a rotary joint that can fulfill the requirement of transmission at mmWave frequencies and rotations up to approx. 1000 rpm (400 km/h with 1 m long arm) at the same time is not available off-the-shelf. Therefore, for mmWave, we transmit a 5.5 GHz intermediate frequency (IF) signal through the rotary joint and the rotating arm. At the end of the rotary arm, we perform mixing to obtain a mmWave frequency signal. Due to the fact that the transmitter needs to rotate at high speeds, it is not possible to mount "bulky equipment" at the end of the rotary arm. At 400 km/h with 1 m arm length the centripetal acceleration is approx. $1250 g$ ($g = 9.81 \frac{m}{s^2}$). Therefore, we designed our own transmitter out of lightweight surface mount device (SMD) components, consisting of an up-converter, a bandpass filter, a power amplifier, a directional coupler, and an antenna. We mount all this on a printed circuit board (PCB) at the end of the rotary arm.

To compare sub 6 GHz and mmWave in the same scenario, we can mount either the sub 6 GHz antenna or the mmWave

transmitter PCB at the end of the rotary arm (see Fig. 1).

Due to reciprocity, for channel sounding, the direction of data transmission is irrelevant. In the sub 6 GHz case, the rotary unit can be used either as a transmitter or receiver, since it consists only of an antenna, cables, and a rotary joint. In the mmWave case, however, we perform frequency conversion at the end of the arm. Therefore we had to decide whether the rotary unit is employed as a transmitter or receiver. Since we had an off-the-shelf Rohde & Schwarz signal analyzer R&S FSW67 available as a receiver, we decided to employ a rotating transmitter. The static receiver is located in the room next to the rotating transmitter.

A. sub 6 GHz

First, a 1 GHz wide data signal intended to be transmitted to the receiver side, is generated off-line using MATLAB. Next, this signal is digitally up-converted to $f_{RF} = 2.55$ GHz by an Arbitrary Waveform Generator (AWG M8195A). The up-converted signal is transmitted to the end of the rotary arm through a rotary joint and cables. Finally, the signal is radiated by an omnidirectional monopole antenna. The omnidirectional antenna is selected to enhance multi-path characteristics compared to a directive antenna which does spatial filtering.

On the receiver side, we use a Rohde & Schwarz signal analyzer (R&S FSW67) with a directive horn antenna to obtain high gain. The received signal is analyzed in MATLAB to obtain channel coefficients.

B. mmWave

In the mmWave case, in addition to an antenna, the transmitter circuit consists of several SMD components,

including an up-converter, a bandpass filter, a power amplifier and a directional coupler. As shown in Fig. 2, the data signal is first up-converted to an Intermediate Frequency (IF) of $f_{IF} = 5.5$ GHz. Besides the IF signal, the rotating transmitter requires a Local Oscillator (LO) tone to perform the up-conversion of the IF signal to a Radio Frequency (RF) range. The LO tone at $f_{LO} = 10$ GHz is generated by a signal generator (R&S SMF). Through two single-channel rotary joints, IF and LO signals are fed to the rotary transmitter circuit.

Using the LO signal, a transceiver chip (MAMF-011024) up-converts the IF signal to form an RF signal at $f_{RF} = 25.5$ GHz

$$f_{RF} = 2f_{LO} + f_{IF}. \quad (1)$$

Besides the desired 25.5 GHz, the up-converter output contains additional spectral products such as higher-order harmonics and intermodulation products. To suppress these products, we apply a 25 – 26 GHz bandpass filter (B259MC1S).

Since the filter output power level is too low to overcome the large path loss inevitable for mmWave frequencies, we use a power amplifier (HMC994AM5E) to increase the transmit power. The power amplifier uses an active bias controller (HMC980LP4E) that can automatically adjust the gate voltage to achieve a constant bias current.

For proper operation, the transceiver and the active bias controller used, require a +5V and a +10.175V DC power supply. Since both rotary joints have already been utilized to transmit high-frequency signals, there are no rotary joints left to transmit the DC supply during the rotation. As a possible solution, we mounted a battery supply near the center of rotation where the centripetal acceleration is low.

If the mmWave transmit signal is directly fed into the antenna, we cannot measure it. Consequently, we cannot measure the channel transfer function by knowing the received signal only. Therefore, we placed a directional coupler (FPC07181) between the power amplifier and the antenna to provide a replica of the transmitter output signal through its coupling path. By knowing both, the replica of the transmitter output signal and the received signal, we can determine the channel transfer function. Similar to the sub 6 GHz case, a directive horn antenna connected to the signal analyzer receives the signal.

C. Synchronization

To provide precise frequency synchronization, the arbitrary waveform generator at the transmitter and the signal analyzer at the receiver are interconnected with a 10 MHz reference. This connection by a cable is a big plus of the proposed setup, not feasible in, for example, drive-by measurements, where expensive rubidium frequency standards need to be employed to ensure precise synchronization.

The same holds true for time synchronization. In the proposed setup, we can repeatedly trigger a measurement at a

precisely defined position of a rotary arm by using a rotational encoder. The signal of the rotary encoder is decoded by a counter and a comparator to form a trigger signal that is fed through cables to the arbitrary waveform generator at the transmitter and the signal analyzer at the receiver.

III. CONCLUSION

We proposed a measurement methodology for performance comparison of sub 6 GHz and mmWave high-mobility scenarios. The proposed testbed is able to perform repeatable measurements in a controlled measuring environment at both, sub 6 GHz and mmWave frequencies at the same channel conditions. Performing repeatable measurements allows us to measure the wireless channels at the same positions for different bands. We are therefore able to conduct comparisons of measured real-world channels on a small scale (in terms of wavelengths in mmWave systems) and a fast-fading scale.

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